Emulating End-to-End Reliable Flow Control over Unreliable Communication Channels

In this assignment you will be building support for end-to-end reliable data transfer on top of an unreliable communication channel, with a window-based flow control approach.

You have been introduced to the function calls socket, bind, sendto, and recvfrom to work with UDP sockets. Assume that the TCP sockets are not there. We know that UDP sockets are not reliable, meaning that there is no guarantee that a message sent using a UDP socket will reach the receiver. We want to implement our own socket type, called KTP socket, that will guarantee that any message sent using a KTP socket is always delivered to the receiver in order. Also, like UDP sockets. KTP is message-oriented, and not byte-oriented.

For each KTP socket, the sender and the receiver maintain a sending window (swnd) and a receiving window (rwnd), respectively. swnd is a data structure that contains the sequence numbers of the messages sent but not yet acknowledged, and the window size indicating the maximum number of messages that can be sent without receiving the corresponding acknowledgement (this should not be more than the available space at the receiver buffer at any instance of time). Similarly, rwnd is a data structure indicating the sequence numbers that the receiver is expecting to receive and a window size indicating the maximum number of messages it can receive depending on the available space at the receiver-side message buffer. The messages are of fixed size (assume that message size is 512 Bytes), and the messages have increasing sequence numbers (sequence number is of 8 bit length). The sender always starts with the sequence number 1. The receiver maintains a message buffer of size equivalent to 10 messages. swnd and rwnd sizes are represented in terms of the number of messages; at any instance of time, rwnd size is equivalent to the amount of free space in the receiver-side message buffer. Once the receiver receives an in-order message, the message is written to the buffer after removing the KTP header, the free space in the buffer is computed and the rwnd size is updated accordingly. The receiver then sends an ACK message to the sender which piggybacks the updated rwnd size, and the sequence number of the last in-order message received within rwnd. If the receiver receives an out of order message, it keeps the message in the buffer (if the message sequence number is within rwnd) but does not send any ACK message. Note that the KTP receiver also needs to handle the duplicate messages by identifying them with the sequence number and then dropping them if already received once. The receiver side application periodically reads the messages (in 512 Bytes block, the entire message is read atomically) from the message buffer and frees it up.

At the sender side, KTP maintains a sender-side message buffer where the sender-side application writes data in a block of 512 Bytes. KTP initially sets swnd = 10 and starts sending the messages after adding the required header (the header contains the message sequence number; think of if you need any additional field in the header). Once the KTP sender receives an ACK message, it updates the swnd accordingly (slides the window till the last message acknowledged and increases/decreases the window size based on the piggybacked rwnd size in the ACK message). The KTP sender also sets a timeout equivalent to T seconds. If it does not receive an ACK message within the timeout duration, it retransmits all the messages from the swnd (which had been transmitted earlier but not been acknowledged).

Multiple user applications can create KTP sockets simultaneously; however, in your implementation, you can assume that at any instance, there can be a maximum of N number of active KTP sockets. To implement these KTP sockets, we use the following:

1. UDP sockets through which all transport layer communication happens for the corresponding KTP socket.
2. An init process that initializes two threads R and S, and a shared memory as mentioned next. Thread R handles all messages received from the UDP socket, and thread S handles the timeouts and retransmissions. More details of R and S are given later.

3. A shared memory chunk SM containing the information about N KTP sockets. SM[i] is a structure indicating (i) whether the KTP socket with ID i is free or allotted, (ii) process ID for the process that created the KTP socket, (iii) mapping from the KTP socket i to the corresponding UDP socket, (iv) the IP and port address of the other end of the KTP socket, (v) the send buffer, and (vi) the receive buffer of the KTP socket with ID i (fixed size array of messages), (vii) swnd (a structure of the sender window size along with an array of message sequence numbers that have been sent but not acknowledged), and (viii) rwnd (a structure of the receiver window size along with an array of message sequence numbers that have been received but not acknowledged).
4. The init process should also start a garbage collector process to clean up the corresponding entry in the KTP socket if the corresponding process is killed and the socket has not been closed explicitly.

You will be implementing a set of function calls k_socket, k_bind, k_sendto, k_recvfrom, and k_close that implement KTP sockets. The parameters and return values to these functions are exactly the same as the corresponding functions of the UDP socket, except for k_bind. The functions will be implemented as a library. Any user wishing to use KTP sockets will write a C program that will call these functions in the same sequence as when using UDP sockets. A brief description of the functions is given below.

- k_socket – This function opens an UDP socket with the socket call. The parameters to these are the same as the normal socket() call, except that it will take only SOCK_KTP as the socket type. k_socket() checks whether a free space is available in the SM, creates the corresponding UDP socket if a free space is available, and initializes SM with corresponding entries. If no free space is available, it returns -1 with the global error variable set to ENOSPACE.

- k_bind – binds the socket with some address-port using the bind call. Bind is necessary for each KTP socket irrespective of whether it is used as a server or a client. This function takes the source IP, the source port, the destination IP and the destination port. It binds the UDP socket with the source IP and source port, and updates the corresponding SM with the destination IP and destination port.

- k_sendto – writes the message to the sender side message buffer if the destination IP/Port matches with the bounded IP/Port as set through k_bind(). If not, it drops the message, returns -1 and sets the global error variable to ENOTBOUND. If there is no space is the send buffer, return -1 and set the global error variable to ENOSPACE. Note that you need to define these error variables properly in a header file.

- k_recvfrom – looks up the receiver-side message buffer to see if any message is already received. If yes, it returns the first message and deletes that message from the table. If not, it returns with -1 and sets a global error variable to ENOMESSAGE, indicating no message has been available in the message buffer.

- k_close– closes the socket and cleans up the corresponding socket entry in the SM and marks the entry as free.

The thread R behaves in the following manner. It waits for a message to come in a recvfrom() call from any of the UDP sockets (you need to use select() to keep on checking whether there is any incoming message on any of the UDP sockets, on timeout check whether a new KTP socket has been created and include it in the read/write set accordingly). When it receives a message, if it is a data message, it stores it in the receiver-side message buffer for the corresponding KTP socket (by searching SM with the IP/Port), and sends an ACK message to the sender. In addition it also sets a flag nospace if the available space at the receive buffer is zero. On a timeout over select() it additionally checks whether the flag nospace was set but now there is space available in the receive buffer. In that case, it sends a duplicate ACK message with the last acknowledged sequence number but with the updated rwnd size, and resets the flag (*there might be a problem here – try to find it out and resolve!*). If the received message is an ACK message in response to a previously sent message, it updates the swnd and removes the message from the sender-side message buffer for the corresponding KTP socket. If the received message is a duplicate ACK message, it just updates the swndsize.

The thread S behaves in the following manner. It sleeps for some time (T/2), and wakes up periodically. On waking up, it first checks whether the message timeout period (T) is over (by computing the time difference between the current time and the time when the messages within the window were sent last) for the messages sent over any of the active KTP sockets. If yes, it retransmits all the messages within the current swnd for that KTP socket. It then checks the current swnd for each of the KTP sockets and determines whether there is a pending message from the sender-side message buffer that can be sent. If so, it sends that message through the UDP sendto() call for the corresponding UDP socket and updates the send timestamp.

Design the message formats and the shared memory properly. Note that different fields in the shared memory are accessed by different threads and processes and would require proper mutual exclusion.

Testing your code:

To test the program, write two programs user1.c and user2.c. user1.c creates an KTP socket M1, binds it to IP_1, Port_1 (local IP and Port), IP_2, Port_2 (remote IP and Port). User2.c creates an KTP socket M2 and binds it to IP_2, Port_2 (local IP and Port), IP_1, Port_1 (remote IP and Port). M1 then uses the k_sendto() call to transfer a large file (size > 100 KB), and M2 uses the k_recvfrom() to receive the contents of that file. You can run user1.c and user2.c with different IP/Port pairs to create multiple KTP sockets.

You should first start with one pair of KTP sockets, i.e. with a single instance of user1.c and user2.c, and then extend it for multiple applications. If your code supports only two statically assigned sockets with the support of all other functionalities, then you'll be evaluated out of 75% of the total marks.

As the actual number of drops in your machine or in the lab environment will be near 0, you will need to simulate an unreliable link. To do this, in the library created, add a function called dropMessage() with the following prototype:

int dropMessage(float p)

where p is a probability between 0 and 1. This function first generates a random number between 0 and 1. If the generated number is < p, then the function returns 1, else it returns
0. Now, in the code for thread R, after a message is received (by the recvfrom() call on the UDP socket), first make a call to dropMessage(). If it returns 1, do not take any action on the message (irrespective of whether it is data or ack) and just return to wait to receive the next message. If it returns 0, continue with the normal processing in R. Thus, if dropMessage() returns 1, the message received is not processed and hence can be thought of as lost. Link the programs in user1.c and user2.c with this new library. Submit your code with the dropMessage() calls in R, do NOT remove these calls from your code before you submit.

The value of T should be 5 seconds (do not hardcode it deep inside your code, specify it in a
.h file (see below)). The value of the parameter p (the probability) should also be specified in the same .h file (see below). When you test your program, vary p from 0.05 to 0.05 in steps of 0.05 (0.05, 0.1, 0.15, 0.2....,0.45, 0.5). For each p value, for the same file, count the average number of transmissions made to send each message (total number of transmissions that are made to send the message / no. of messages generated from the file). Report this in a table in the beginning of the file documentation.txt (see below).

Even though it is not needed for this mini project, you should try to also vary T (especially low values) and see its effect. This is an extremely important parameter that affects the transmission. If you do this, set p to 0, and use the nanosleep() call of Linux in S as the sleep() call has a resolution of seconds only, so you cannot set T to anything less than 1 second using a sleep() call.

What to submit:

The five required functions, plus the function dropMessage(), should be implemented as a static library called libksocket.a so that a user can write a C program using these calls for reliable communication and link with it (the function dropMessage() will not be called by the user but you will call it when you test your program). Look up the ar command under Linux to see how to build a static library. Building a library means creating a .h file containing all definitions needed for the library (for ex., among other things, you will #define SOCK_KTP

here), and a .c file containing the code for the functions in the library. This .c file should not contain any main() function, and will be compiled using ar command to create a .a library file. Thus, you will write the .h header file (name it ksocket.h) and the .c file (name it ksocket.c) from which the .a library file will be generated. Any application wishing to call these functions will include the .h file and link with the libksocket.a library. For example, when we want to use functions in the math library like sqrt(), we include math.h in our C file, and then link with the math library libm.a.

The value of the parameter T should be #defined as "#define T 5" in the ksocket.h file. The value of the parameter p should also be #defined in this .h file.

You should submit the following files in a single tar.gz file:

- ksocket.h and ksocket.c
- initksocket.c that initializes the KTP socket by starting the R and S threads and the garbage collector process
- user1.c and user2.c
- a makefile to generate libksocket.a
- a makefile to create the executable to run initksocket.c
- a makefile to create the two executable files to run from user1.c and user2,c respectively
- a file called documentation.txt containing the following:
    - For ksocket.h and ksocket.c, as well as for initksocket.c, give a list of all data structures used and a brief description of their fields, and a list of all functions along with what they do in ksocket.c and initksocket.c.
    - The table with the results for varying p values as described earlier.